

RAFFLES INSTITUTION

Higher 1

CANDIDATE
NAME

--

CIVICS
GROUP

1	1	S0	3	
---	---	----	---	--

INDEX
NUMBER

1	1				
---	---	--	--	--	--



BIOLOGY

8875/02

Paper 2 Core Paper

12 September 2011

2 hours

Additional materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your index number, CT group & name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer all questions.

Section B

Answer only **one** question.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
SECTION A	
1	/11
2	/10
3	/11
4	/8
SECTION B	
5 or 6	/20
TOTAL	/60

This document consists of **9** printed pages.



Raffles Institution
Internal Examination

Section AAnswer **all** questions in this section.FOR
EXAMINER'S
USE

- 1 The main function of a chromosome is to act as a **template** for the synthesis of RNA molecules, since only in this way does the genetic information stored in chromosomes become directly useful to the cell. RNA synthesis is a highly selective process.

(a) Explain what is meant by the term **template**. [3]

Fig.1.1 is a section through the rough endoplasmic reticulum representing the synthesis of proteins

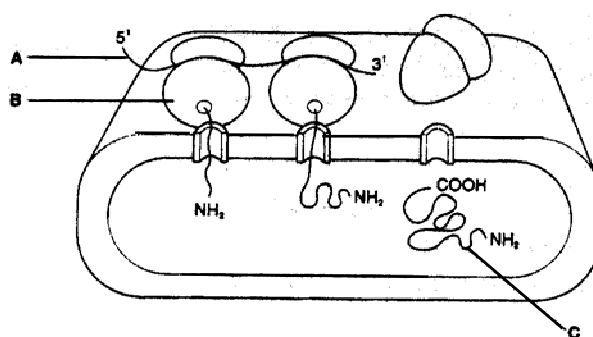


Fig. 1.1

(b) Describe and explain the role in protein synthesis of

(i) structure **A**; [3]

(ii) structure **B**. [3]

FOR
EXAMINER'S
USE

(c) Explain why mutations in the coding region for **C** can still result in identical polypeptide **C**? [2]

[Total: 11m]

2 (a) Explain how a mutation that results in the substitution of a single base may affect the phenotype of an organism. [2]

(b) The human disorder *incontinentia pigmenti* is a rare syndrome in which melanin is not retained by melanocytes, causing swirling pigment lines on the skin. Su Lin, a woman with the disorder, marries Hock Seng, a normal man. She gives birth to three normal daughters, two affected daughters, and two normal sons. In addition, she had three miscarriages of deformed male fetuses. (Assume that the syndrome is controlled by a single gene).

(i) State the pattern of inheritance of *incontinentia pigmenti*. [2]

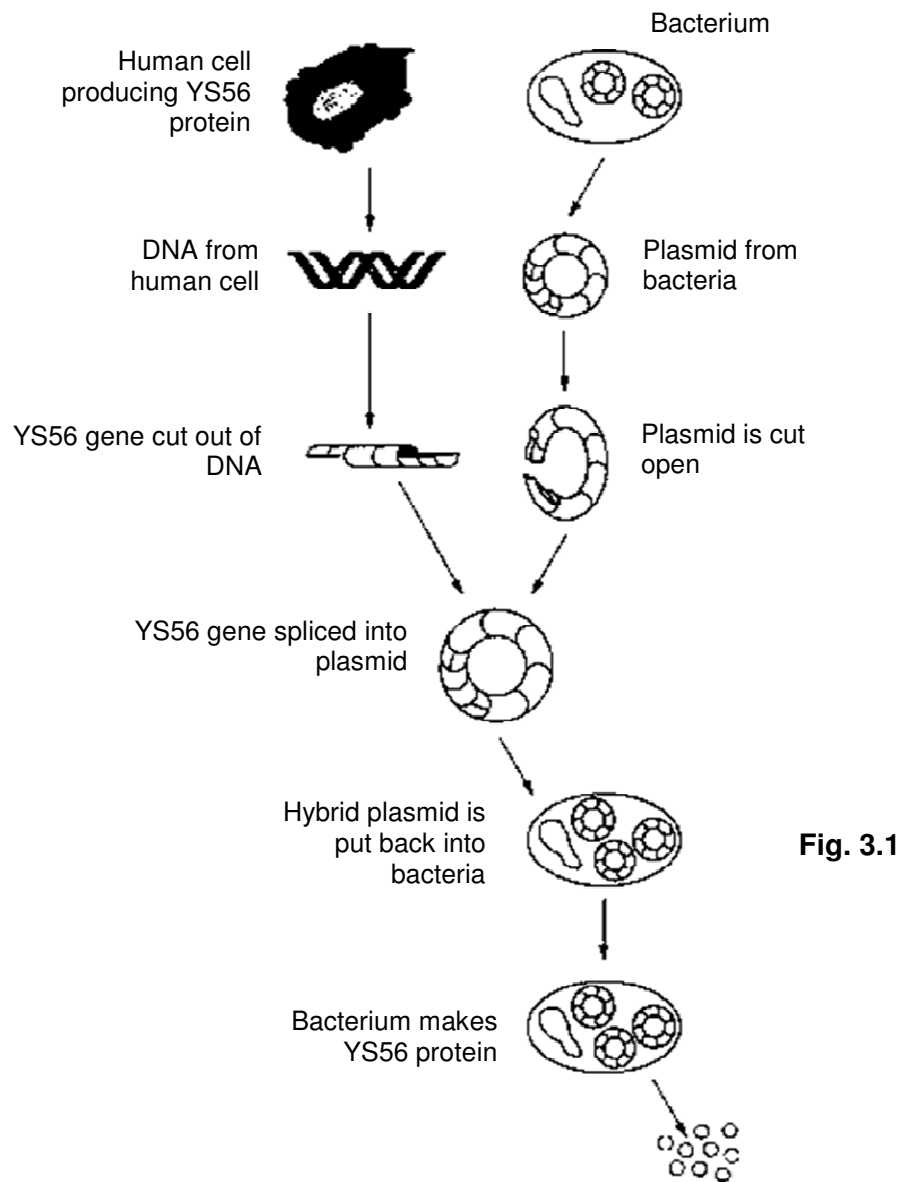
- (ii) What is the probability that the next child that Su Lin and Hock Seng have is a healthy and normal female. Explain your answer in the form of a genetic diagram. [6]

FOR
EXAMINER'S
USE

Probability of Su Lin and Hock Seng having a healthy normal female child is: _____

[Total: 10m]

3 **Fig. 3.1** below illustrates how a human protein, YS56, can be produced using bacteria.



(a) Restriction enzymes are used to cut the YS56 gene out from human DNA. Explain the mechanism of action of restriction enzymes. [3]

- (b) State the enzyme that is used to splice the YS56 gene into the plasmid, and describe the reaction it catalyses. [2]

FOR
EXAMINER'S
USE

- (c) Explain how the recombinant plasmid can be “put back” into the bacteria. [2]

- (d) Describe the properties of a plasmid that make it suitable for use as a cloning vector. [4]

[Total: 11m]

4 Wheat, maize and sorghum are three of the most important cereal crops in the world.

(a) **Fig. 4.1** shows the effect of temperature on the rate of photosynthesis of wheat plants.

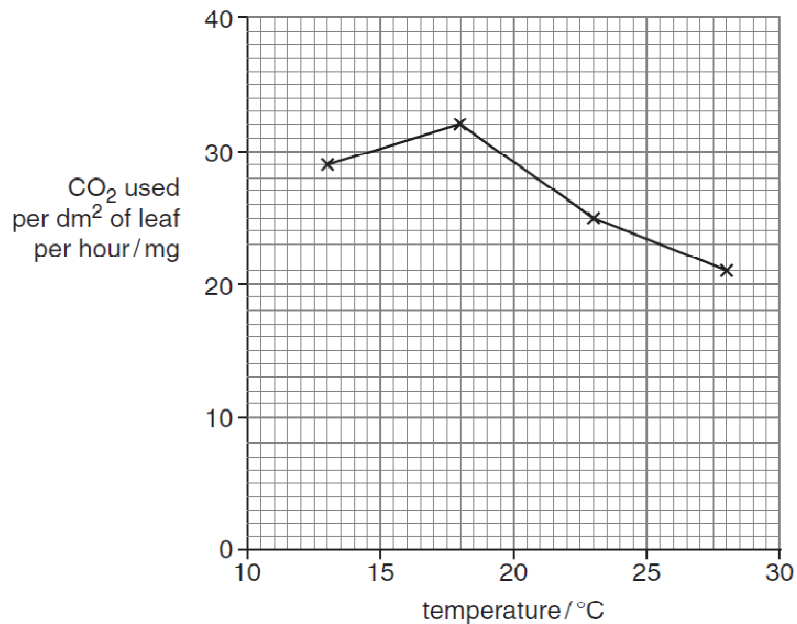


Fig. 4.1

(i) With reference to **Fig. 4.1**, describe the effect of temperature on the rate of photosynthesis of wheat plants. [2]

(ii) Suggest why temperature affects the rate of photosynthesis in the way you have described in (i). [2]

- (b) The conditions in which young plants of wheat and maize are grown affect their ability to photosynthesise at high and low temperatures when they are mature. Young maize and wheat plants were grown to maturity at high and low temperatures. When they were mature, their rate of photosynthesis was measured at different temperatures. The results are shown in **Fig. 4.2**.

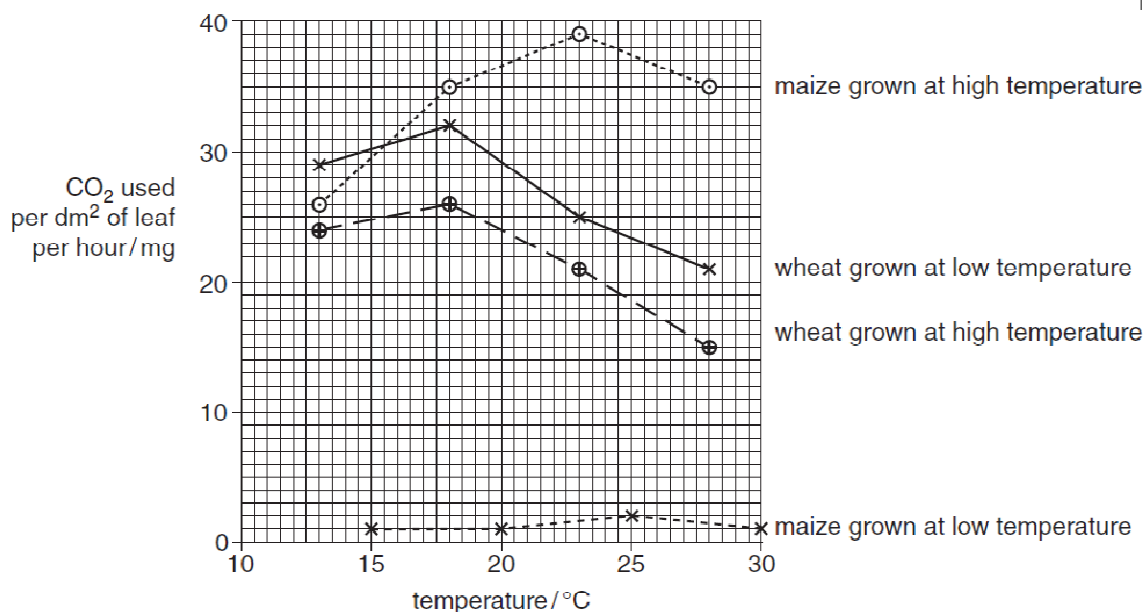


Fig. 4.2

- (i) With reference to **Fig. 4.2**, compare the effect of temperature on the rate of photosynthesis of maize plants and wheat plants that were grown at a high temperature when they were young. [2]

- (ii) Low temperatures slow down the formation of the membranes inside chloroplasts in maize leaves, but not in wheat leaves. Use this information to explain the differences between the results for maize and wheat grown at low temperatures, shown in **Fig. 4.2**. [2]

[Total: 8m]

Section B

Answer **ONE** question.

FOR
EXAMINER'S
USE

Write your answer to this question on separate answer paper provided.
Your answer should be illustrated by large, clearly labeled diagrams, where appropriate.
Your answer must be in continuous prose.
Your answer must be set out in sections **(a)** and **(b)**, as indicated in the question.

- 5** (a) Describe the roles played by proteins in the plasma membranes [8]
- (b) Describe the main ways in which a globular protein differs from DNA. [7]
- (c) Explain the effects of pH on the rate of enzyme-catalysed reactions. [5]
- 6** (a) Explain, with examples, how environmental factors act as forces of natural selection. [8]
- (b) Describe the polymerase chain reaction (PCR). [6]
- (c) Describe the unique features of stem cells. [6]

---- End of paper ----